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Integrated circuit package and method

Abstract

In an embodiment, a device includes: a processor die including circuit blocks, the circuit blocks including active devices of a first technology node; a power gating die including power semiconductor devices of a second technology node, the second technology node larger than the first technology node; and a first redistribution structure including first metallization patterns, the first metallization patterns including power supply source lines and power supply ground lines, where a first subset of the circuit blocks is electrically coupled to the power supply source lines and the power supply ground lines through the power semiconductor devices, and a second subset of the circuit blocks is permanently electrically coupled to the power supply source lines and the power supply ground lines.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a divisional of U.S. patent application Ser. No. 16/882,132, filed May 22, 2020, entitled “Integrated Circuit Package and Method,” which claims the benefit of U.S. Provisional Application No. 62/916,954, filed on Oct. 18, 2019, which applications are hereby incorporated herein by reference.

BACKGROUND

(1) As semiconductor technologies continue to evolve, integrated circuit dies are becoming increasingly smaller. Further, more functions are being integrated into the dies. Accordingly, the numbers of input/output (I/O) pads needed by dies has increased while the area available for the I/O pads has decreased. The density of the I/O pads has risen quickly over time, increasing the difficulty of die packaging. Some applications call for greater parallel processing capabilities of integrated circuit dies. Packaging technologies may be used to integrate of multiple dies, allowing a greater degree of parallel processing capabilities.

(2) In some packaging technologies, integrated circuit dies are singulated from wafers before they are packaged. An advantageous feature of this packaging technology is the possibility of forming fan-out packages, which allow the I/O pads on a die to be redistributed to a greater area. The number of I/O pads on the surfaces of the dies may thus be increased.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1, 2, and 3 are cross-sectional views of intermediate steps during a process for forming an integrated circuit package, in accordance with some embodiments.

(3) FIGS. 4, 5, and 6 are cross-sectional views of intermediate steps during a process for forming a system implementing an integrated circuit package, in accordance with some embodiments.

(4) FIG. 7 is a cross-sectional view of an integrated circuit package and a system implementing the integrated circuit package, in accordance with some other embodiments.

- (5) FIGS. **8, 9, 10, 11, 12, 13, 14, 15,** and **16** are cross-sectional views of intermediate steps during a process for forming a system implementing an integrated circuit package, in accordance with some embodiments.
- (6) FIG. **17** is a cross-sectional view of an integrated circuit package and a system implementing the integrated circuit package, in accordance with some other embodiments.
- (7) FIG. **18** is a cross-sectional view of an integrated circuit package and a system implementing the integrated circuit package, in accordance with some other embodiments.
- (8) FIG. **19** is a cross-sectional view of an integrated circuit package and a system implementing the integrated circuit package, in accordance with some other embodiments.
- (9) FIGS. **20, 21, 22, 23,** and **24** are cross-sectional views of intermediate steps during a process for forming an integrated circuit package, in accordance with some embodiments.
- (10) FIGS. **25, 26, 27,** and **28** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package, in accordance with some embodiments.
- (11) FIGS. **29, 30, 31, 32, 33,** and **34** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package, in accordance with some other embodiments.
- (12) FIGS. **35, 36, 37, 38, 39,** and **40** are cross-sectional views of intermediate steps during a process for forming an integrated circuit package, in accordance with some other embodiments.
- (13) FIG. **41** is a cross-sectional view of an integrated circuit package, in accordance with some other embodiments.

DETAILED DESCRIPTION

- (14) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (15) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (16) In accordance with some embodiments, integrated circuit packages are formed having processor devices and power gating devices that are separate semiconductor devices. The power gating devices are formed with larger active devices than the processor devices, and include power semiconductor devices providing power gating to the processor devices. Switch transistors of a large technology node may thus be used for power delivery, allowing a reduction in the power consumption of the resulting integrated circuit packages.
- (17) FIGS. **1, 2,** and **3** are cross-sectional views of intermediate steps during a process for forming an integrated circuit package **100**, in accordance with some embodiments. The integrated circuit package **100** is formed by stacking semiconductor devices on a wafer **102**. Stacking of devices in one device region **102A** of the wafer **102** is illustrated, but it should be appreciated that the wafer

102 may have any number of device regions, and semiconductor devices may be stacked to form an integrated circuit package in each device region. The semiconductor devices can be bare integrated circuit dies or packaged dies. In the embodiment illustrated, each semiconductor device is a bare integrated circuit die. In other embodiments, one or more of the illustrated semiconductor devices can be packaged dies that are encapsulated.

(18) In FIG. **1**, the wafer **102** is obtained. The wafer **102** comprises a processor device **10** in the device region **102A**. The processor device **10** will be singulated in subsequent processing to be included in the integrated circuit package **100**. The processor device **10** can be any acceptable processor or logic device, such as a central processing unit (CPU), graphics processing unit (GPU), arithmetic logic unit (ALU), system-on-a-chip (SoC), application processor (AP), image signal processor (ISP), digital signal processing (DSP), field programmable gate array (FPGA), microcontroller, artificial intelligence (AI) accelerator, or the like. As discussed further below, the processor device **10** is formed without power gating features.

(19) The processor device **10** may be processed according to applicable manufacturing processes to form integrated circuits. For example, the processor device **10** includes a semiconductor substrate **12**, such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **12** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The semiconductor substrate **12** has an active surface **12A** and an inactive surface **12N**.

(20) Devices may be formed at the active surface **12A** of the semiconductor substrate **12**. The devices may be active devices (e.g., transistors, diodes, etc.), capacitors, resistors, etc. The inactive surface **12N** may be free from devices. An inter-layer dielectric (ILD) is over the active surface **12A** of the semiconductor substrate **12**. The ILD surrounds and may cover the devices. The ILD may include one or more dielectric layers formed of materials such as Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), undoped Silicate Glass (USG), or the like.

(21) An interconnect structure **14** is over the active surface **12A** of the semiconductor substrate **12**. The interconnect structure **14** interconnects the devices at the active surface **12A** of the semiconductor substrate **12** to form an integrated circuit. The interconnect structure **14** may be formed by, for example, metallization patterns in dielectric layers. The metallization patterns include metal lines and vias formed in one or more dielectric layers. The metallization patterns of the interconnect structure **14** are electrically coupled to the devices at the active surface **12A** of the semiconductor substrate **12**.

(22) Die connectors **16** are at a front side **10F** of the processor device **10**. The die connectors **16** may be conductive pillars, pads, or the like, to which external connections are made. The die connectors **16** are in and/or on the interconnect structure **14**. The die connectors **16** can be formed of a metal, such as copper, aluminum, or the like, and can be formed by, for example, plating, or the like.

(23) A dielectric layer **18** is at the front side **10F** the processor device **10**. The dielectric layer **18** is in and/or on the interconnect structure **14**. The dielectric layer **18** laterally encapsulates the die connectors **16**, and the dielectric layer **18** is laterally coterminous with sidewalls of the processor device **10**. Initially, the dielectric layer **18** may bury the die connectors **16**, such that the topmost surface of the dielectric layer **18** is above the topmost surfaces of the die connectors **16**. The dielectric layer **18** may be a polymer such as polybenzoxazole (PBO), polyimide, a benzocyclobuten (BCB) based polymer, or the like; a nitride such as silicon nitride or the like; an oxide such as silicon oxide, PSG, BSG, BPSG, or the like; the like, or a combination thereof. The dielectric layer **18** may be formed, for example, by spin coating, lamination, chemical vapor

deposition (CVD), or the like. After formation, the die connectors **16** and dielectric layer **18** can be planarized using, e.g., a chemical-mechanical polish (CMP) process, an etch back process, the like, or combinations thereof. After planarization, surfaces of the die connectors **16** and dielectric layer **18** are planar and are exposed at the front side **10F** of the processor device **10**.

(24) The processor device **10** includes a plurality of circuit blocks. A circuit block is a logical block or unit of circuits of the processor device **10**. In other words, a circuit block comprises a subset of the circuits of the processor device **10**, where the circuits in a circuit block are all related to the same domain. Examples of circuit blocks include arithmetic circuit blocks, memory circuit blocks, DSP circuit blocks, and the like.

(25) A power gating device **20** is also obtained. The power gating device **20** is part of the power delivery network for the processor device **10**. Specifically, the power gating device **20** provides power gating to some or all of the circuit blocks of the processor device **10**. The power gating device **20** is operable at runtime to receive control signals and turn the circuit blocks of the processor device **10** on or off responsive to the control signals. For example, the circuit blocks of the processor device **10** can be turned on when needed and can be turned off when not in use, thus reducing the leakage power of the unused circuit blocks. The power consumption of the processor device **10** may thus be reduced. Further, forming the processor device **10** without power gating features allows for a reduction in the area of the processor device **10**. For example, moving the power gating features to a dedicated power gating device **20** allows the area of the processor device **10** to be reduced by to 20%.

(26) The power gating device **20** may be processed according to applicable manufacturing processes to form integrated circuits. For example, the power gating device **20** includes a semiconductor substrate **22**, an interconnect structure **24**, die connectors **26**, and a dielectric layer **28** which, respectively, can be similar to the semiconductor substrate **12**, interconnect structure **14**, die connectors **16**, and dielectric layer **18**. The die connectors **26** and dielectric layer **28** are exposed at a front side **20F** of the power gating device **20**. The power gating device **20** further includes conductive vias **30**, which are formed extending into the semiconductor substrate **22**. The conductive vias **30** are electrically coupled to metallization patterns of the interconnect structure **24**.

(27) As an example to form the conductive vias **30**, recesses can be formed in the interconnect structure **24** and/or semiconductor substrate **22** by, for example, etching, milling, laser techniques, a combination thereof, and/or the like. A thin dielectric material may be formed in the recesses, such as by using an oxidation technique. A thin barrier layer may be conformally deposited in the openings, such as by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), thermal oxidation, a combination thereof, and/or the like. The barrier layer may be formed from an oxide, a nitride, or an oxynitride, such as titanium nitride, titanium oxynitride, tantalum nitride, tantalum oxynitride, tungsten nitride, a combination thereof, and/or the like. A conductive material may be deposited over the barrier layer and in the openings. The conductive material may be formed by an electro-chemical plating process, CVD, ALD, PVD, a combination thereof, and/or the like. Examples of conductive materials are copper, tungsten, aluminum, silver, gold, a combination thereof, and/or the like. Excess conductive material and barrier layer is removed from the surface of the interconnect structure **24** and/or semiconductor substrate **22** by, for example, a CMP. Remaining portions of the barrier layer and conductive material form the conductive vias **30**.

(28) In the embodiment illustrated, the conductive vias **30** are not yet exposed at a back side **20B** of the power gating device **20**. Rather, the conductive vias **30** are buried in the semiconductor substrate **22**. As discussed further below, the conductive vias **30** will be exposed at the back side **20B** of the power gating device **20** through a planarization process in subsequent processing. After exposure, the conductive vias **30** can be referred to as through-substrate vias or through-silicon vias (TSVs).

(29) The power gating device **20** includes power semiconductor devices (e.g., semiconductor

devices used for power electronics) to provide power gating to the processor device **10**. Specifically, the power gating device **20** can include switch transistors, power gating controllers, and a power gating fabric. The power semiconductor devices of the power gating device **20** can be bipolar devices, such as insulated-gate bipolar transistor (IGBT) devices or the like, or can be power MOSFET devices, such as complementary metal-oxide-semiconductor (CMOS) devices, double diffused metal-oxide-semiconductor (DMOS) devices, or the like. In some embodiments, the processor device **10** and power gating device **20** are formed by different semiconductor process technologies, and have devices (e.g., active devices) of different pitches. For example, the processor device **10** can comprise CMOS devices, and the power gating device **20** can comprise IGBT devices or DMOS devices. The active surface of the semiconductor substrate **22** includes switching device regions **22S** comprising switch transistors. The switch transistors of the switching device regions **22S** are connected to power supply source (V.sub.DD) lines (discussed further below) and, when activated, electrically couple the V.sub.DD lines to the circuit blocks of the processor device **10**. Likewise, the metallization patterns of the interconnect structure **24** can form some or all of the power gating fabric. For example, the interconnect structure **24** can include metal lines and vias that are used to route the V.sub.DD lines from the switch transistors to the circuit blocks of the processor device **10**.

(30) The metallization patterns of the interconnect structure **24** can form some, all, or none of the power gating fabric. In some embodiments, the interconnect structure **24** of the power gating device **20** comprises all of the power gating fabric. In some embodiments, the interconnect structure **24** of the power gating device **20** comprises some of the power gating fabric, and the interconnect structure **14** of the processor device **10** comprises some of the power gating fabric. In some embodiments, the interconnect structure **24** of the power gating device **20** does not comprise the power gating fabric, and the interconnect structure **14** of the processor device **10** comprises all of the power gating fabric.

(31) The power delivery performance of the switch transistors can significantly impact the overall power delivery performance of the power gating device **20**. Specifically, switch transistors of a large technology node can accommodate greater rush currents and have a smaller voltage drop (e.g., IR drop) when the circuit blocks of the processor device **10** are turned on. Similarly, switch transistors of a large technology node can have a greater threshold voltage, which allows the switch transistors to have less power leakage. In accordance with some embodiment, the processor device **10** have active devices of a smaller technology node (e.g., smaller device pitch or spacing), which are unsuitable for switch transistors but are suitable for logic devices, and the power gating device **20** have active devices of a larger technology node (e.g., larger device pitch or spacing), which are unsuitable for logic devices but are suitable for switch transistors. For example, the processor device **10** can have active devices (e.g., forming circuit blocks) of a technology node in the range of about 2 nm to about 28 nm, and the power gating device **20** can have active devices (e.g., power semiconductor devices) of a technology node in the range of about 3 nm to about 90 nm. Forming the power gating device **20** with larger active devices than the processor device **10** allows the processor device **10** to be scaled down to smaller technology nodes that provide improved logic device performance, while still allowing the power gating device **20** to provide sufficient power delivery performance. Further, forming the power gating device **20** with larger active devices allows the manufacturing costs of the power gating device **20** to be reduced.

(32) In FIG. 2, the power gating device **20** is bonded to the processor device **10** (e.g., the wafer **102**). The processor device **10** and power gating device **20** are directly bonded in a face-to-face manner by hybrid bonding, such that the front side **10F** of the processor device **10** is bonded to the front side **20F** of the power gating device **20**. Specifically, the dielectric layer **18** of the processor device **10** is bonded to the dielectric layer **28** of the power gating device **20** through dielectric-to-dielectric bonding, without using any adhesive material (e.g., die attach film), and a subset of the die connectors **16A** of the processor device **10** are bonded to the die connectors **26** of the power

gating device **20** through metal-to-metal bonding, without using any eutectic material (e.g., solder). The bonding may include a pre-bonding and an annealing. During the pre-bonding, a small pressing force is applied to press the power gating device **20** against the processor device **10** (e.g., the wafer **102**). The pre-bonding is performed at a low temperature, and after the pre-bonding, the dielectric layers **18** and **28** are bonded to each other. The bonding strength is then improved in a subsequent annealing step, in which the dielectric layers **18** and **28** are annealed at a high temperature. After the annealing, bonds, such as fusions bonds, are formed bonding the dielectric layers **18** and **28**. For example, the bonds can be covalent bonds between the material of the dielectric layer **18** and the material of the dielectric layer **28**. The die connectors **16A** and **26** are connected to each other with a one-to-one correspondence. The die connectors **16A** and **26** may be in physical contact after the pre-bonding, or may expand to be brought into physical contact during the annealing. Further, during the annealing, the material of the die connectors **16A** and **26** (e.g., copper) intermingles, so that metal-to-metal bonds are also formed. Hence, the resulting bonds between the processor device **10** and power gating device **20** are hybrid bonds that include both dielectric-to-dielectric bonds and metal-to-metal bonds.

(33) After hybrid bonding, the processor device **10** is electrically coupled to the power gating device **20** by direct bonds, over which control signaling and power/ground signaling is performed. Specifically, the processor device **10** may send control signals to the power gating device **20** over the direct bonds, and may receive power/ground signals from the power gating device **20** over the direct bonds. The latency of signaling and the interconnection bandwidth between the processor device **10** and the power gating device **20** may be improved by the use of direct bonds. Further, the impedance and thus power consumption of the interconnections may also be reduced. As noted above, the metallization patterns of the interconnect structure **24** can form some or all of the power gating fabric. In such embodiments, the power gating fabric is thus connected to the processor device **10** by direct bonds, which may allow for simpler routing than other types of interconnections. In the embodiment shown, one power gating device **20** is bonded to the processor device **10**. In another embodiment, multiple power gating devices **20** are bonded to the processor device **10**.

(34) In FIG. 3, a dielectric layer **104** is formed surrounding the power gating device **20**. The dielectric layer **104** can be formed after placement of the power gating device **20** but before annealing to complete the hybrid bonding, or can be formed after annealing. The dielectric layer **104** fills gaps between the power gating device **20** and other devices in adjacent device regions (not shown) of the wafer **102**, thus protecting the semiconductor devices. The dielectric layer **104** may be an oxide such as silicon oxide, PSG, BSG, BPSG, or the like; a nitride such as silicon nitride or the like; a polymer such as PBO, polyimide, a BCB-based polymer, or the like; an encapsulant such as a molding compound, epoxy, or the like; the like, or a combination thereof. In some embodiments, the dielectric layer **104** is an oxide such as silicon oxide. In some embodiments, the dielectric layer **104** is deposited using a CVD process or the like.

(35) Conductive vias **106** are then formed to extend through the dielectric layer **104**. As an example to form the conductive vias **106**, openings are patterned in the dielectric layer **104**. The patterning may be by an acceptable process, such as by exposing the dielectric layer **104** to light when the dielectric layer **104** is a photo-sensitive material, or by etching the dielectric layer **104** using, for example, an anisotropic etch. The openings expose a subset of the die connectors **16B** of the processor device **10**. A seed layer is formed on the dielectric layer **104** and on portions of the die connectors **16B** exposed by the openings. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In a particular embodiment, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A conductive material is formed on the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may

comprise a metal, such as copper, titanium, tungsten, aluminum, or the like. Excess portions of the seed layer and conductive material are then removed, with the excess portions being portions overlying the dielectric layer **104**. The removal may be by a planarization process. The planarization process is performed on the seed layer, conductive material, dielectric layer **104**, and power gating device **20**. The removal simultaneously removes excess portions of the seed layer and conductive material and exposes the conductive vias **30**. The planarization process may be, for example, a CMP process, a grinding process, an etch back process, the like, or combinations thereof. The remaining portions of the seed layer and conductive material in the openings form the conductive vias **106**. Top surfaces of the dielectric layer **104**, conductive vias **106**, semiconductor substrate **22**, and conductive vias **30** are planar after the planarization process.

(36) FIGS. **4**, **5**, and **6** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package **100**, in accordance with some embodiments. In this embodiment, the integrated circuit package **100** is further processed to include a redistribution structure, and is directly mounted to a package substrate.

(37) In FIG. **4**, a redistribution structure **108** is formed on the dielectric layer **104**, conductive vias **106**, and power gating device **20**. The redistribution structure **108** includes dielectric layers **110** and metallization patterns **112** (sometimes referred to as redistribution layers or redistribution lines) among the dielectric layers **110**. For example, the redistribution structure **108** may include a plurality of metallization patterns **112** separated from each other by respective dielectric layers **110**. The metallization patterns **112** of the redistribution structure **108** are connected to the conductive vias **106** and the power gating device **20** (e.g., to the conductive vias **30**).

(38) In some embodiments, the dielectric layers **110** are formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, a BCB-based polymer, or the like, may be patterned using a lithography mask. In other embodiments, the dielectric layers **110** are formed of a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG; or the like. The dielectric layers **110** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. After each dielectric layer **110** is formed, it is then patterned to expose underlying conductive features, such as portions of the underlying metallization patterns **112**. The patterning may be by an acceptable process, such as by exposing the dielectrics layers to light when the dielectric layers **110** are a photo-sensitive material, or by etching using, for example, an anisotropic etch. If the dielectric layers **110** are photo-sensitive materials, the dielectric layers **110** can be developed after the exposure.

(39) The metallization patterns **112** each include conductive vias and/or conductive lines. The conductive vias extend through the dielectric layers **110**, and the conductive lines extend along the dielectric layers **110**. As an example to form a metallization pattern, a seed layer (not illustrated) is formed over the respective underlying features. For example, the seed layer can be formed on the dielectric layer **104**, conductive vias **106**, semiconductor substrate **22**, and conductive vias **30** when the bottommost level of the redistribution structure **108** is formed, or the seed layer can be formed on a respective dielectric layer **110** and in the openings through the respective dielectric layer **110** when the intermediate/topmost levels of the redistribution structure **108** are formed. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using a deposition process, such as PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization pattern. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal or a metal alloy, such as copper,

titanium, tungsten, aluminum, the like, or combinations thereof. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the metallization pattern for one level of the redistribution structure **108**.

(40) The redistribution structure **108** is illustrated as an example. More or fewer dielectric layers **110** and metallization patterns **112** than illustrated may be formed in the redistribution structure **108** by repeating or omitting the steps described above.

(41) The metallization patterns **112** of the redistribution structure **108** include power supply source (V.sub.DD) lines and power supply ground (V.sub.SS) lines. The V.sub.DD and V.sub.SS lines are connected to the conductive vias **30** and **106**. The switching device regions **22S** of the power gating device **20** are thus electrically coupled to the V.sub.DD and V.sub.SS lines by the conductive vias **30**. A first subset of the circuit blocks of the processor device **10** are electrically coupled to the V.sub.DD and V.sub.SS lines through the power gating device **20**. The first subset of the circuit blocks can thus be turned on and off by the power gating device **20**. A second subset of the circuit blocks of the processor device **10** are electrically coupled to the V.sub.DD and V.sub.SS lines through the conductive vias **106**, bypassing the power gating device **20**. The second subset of the circuit blocks are thus permanently electrically coupled to power and ground. Circuit blocks that are permanently electrically coupled to power and ground are always turned on, and are not power gated.

(42) The metallization patterns **112** of the redistribution structure **108** also include data signal lines, which are electrically coupled to the processor device **10** by the conductive vias **106**. For example, some of the conductive vias **106** couple input/output (I/O) connections of the processor device **10** to the redistribution structure **108**. The processor device **10** may thus be coupled to external devices.

(43) In FIG. 5, conductive connectors **114** are formed connected to the metallization patterns **112** of the redistribution structure **108**. The top dielectric layer **110** of the redistribution structure **108** may be patterned to expose portions of the underlying metallization patterns **112**. In some embodiments, under bump metallurgies (UBMs) may be formed in the openings. The conductive connectors **114** are formed on the UBMs. The conductive connectors **114** may be ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **114** may be formed of a metal or metal alloy, such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **114** are formed by initially forming a layer of solder through such commonly used methods such as evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **114** are metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls.

(44) In FIG. 6, a singulation process is performed by sawing along scribe line regions, e.g., around the device region **102A**. The singulation process includes sawing the redistribution structure **108**, dielectric layer **104**, and wafer **102**. The singulation process separates the device region **102A** (comprising the processor device **10**) from adjacent device regions (not illustrated) of the wafer **102** to form an integrated circuit package **100** comprising the processor device **10**. As noted above, the power gating device **20** is directly bonded to the processor device **10** in a face-to-face manner, without the use of solder. The resulting integrated circuit package **100** is thus free from solder.

After singulation, the redistribution structure **108** (e.g., the dielectric layers **110**), the dielectric layer **104**, and the processor device **10** are laterally coterminous.

(45) The integrated circuit package **100** is then flipped and attached to a package substrate **200** using the conductive connectors **114**. The package substrate **200** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations of these, and the like, may also be used. Additionally, the package substrate **200** may be a SOI substrate. Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, SOI, SGOI, or combinations thereof. The package substrate **200** is, in one alternative embodiment, based on an insulating core such as a fiberglass reinforced resin core. One example core material is fiberglass resin such as FR4. Alternatives for the core material include bismaleimide-triazine (BT) resin, or alternatively, other printed circuit board (PCB) materials or films. Build up films such as Ajinomoto build-up film (ABF) or other laminates may be used for package substrate **200**.

(46) The package substrate **200** may include active and passive devices (not illustrated). Devices such as transistors, capacitors, resistors, combinations of these, and the like may be used to generate the structural and functional requirements of the design for the system. The devices may be formed using any suitable methods.

(47) The package substrate **200** may also include metallization layers and vias (not illustrated) and bond pads **202** over the metallization layers and vias. The metallization layers may be formed over the active and passive devices and are designed to connect the various devices to form functional circuitry. The metallization layers may be formed of alternating layers of dielectric (e.g., low-k dielectric material) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material and may be formed through any suitable process (such as deposition, damascene, dual damascene, or the like). In some embodiments, the package substrate **200** is substantially free of active and passive devices.

(48) The conductive connectors **114** are reflowed to attach the UBMs of the redistribution structure **108** to the bond pads **202**. The conductive connectors **114** connect the package substrate **200**, including metallization layers in the package substrate **200**, to the integrated circuit package **100**, including metallization patterns **112** of the redistribution structure **108**. In some embodiments, passive devices (e.g., surface mount devices (SMDs), not illustrated) may be attached to the integrated circuit package **100** (e.g., bonded to the bond pads **202**) prior to mounting on the package substrate **200**. In such embodiments, the passive devices may be bonded to a same surface of the integrated circuit package **100** as the conductive connectors **114**. In some embodiments, passive devices (e.g., SMDs, not illustrated) may be attached to the package substrate **200**, e.g., to the bond pads **202**.

(49) The conductive connectors **114** may have an epoxy flux (not illustrated) formed thereon before they are reflowed with at least some of the epoxy portion of the epoxy flux remaining after the integrated circuit package **100** is attached to the package substrate **200**. This remaining epoxy portion may act as an underfill to reduce stress and protect the joints resulting from the reflowing the conductive connectors **114**. In some embodiments, an underfill (not illustrated) may be formed between the integrated circuit package **100** and the package substrate **200**, surrounding the conductive connectors **114**. The underfill may be formed by a capillary flow process after the integrated circuit package **100** is attached or may be formed by a suitable deposition method before the integrated circuit package **100** is attached.

(50) FIG. 7 is a cross-sectional view of an integrated circuit package **100** and a system implementing the integrated circuit package **100**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. 6, except the processor device **10** and power gating device **20** are directly bonded in a face-to-back manner. The

metallization patterns **112** of the redistribution structure **108** (e.g., V.sub.DD and V.sub.SS lines) are thus connected to the die connectors **26**, and are electrically coupled to the processor device **10** by the conductive vias **30**.

(51) The processor device **10** and power gating device **20** are directly bonded in a face-to-back manner by hybrid bonding, such that the front side **10F** of the processor device **10** is bonded to the back side **20B** of the power gating device **20**. Specifically, the dielectric layer **18** of the processor device **10** is bonded to the semiconductor substrate **22** of the power gating device **20** through dielectric-to-dielectric bonding, without using any adhesive material (e.g., die attach film), and a subset of the die connectors **16A** of the processor device **10** are bonded to the conductive vias **30** of the power gating device **20** through metal-to-metal bonding, without using any eutectic material (e.g., solder). The bonding may include a pre-bonding and an annealing. During the pre-bonding, a small pressing force is applied to press the power gating device **20** against the processor device **10**. The pre-bonding is performed at a low temperature, and after the pre-bonding, the dielectric layer **18** and semiconductor substrate **22** are bonded to each other. In some embodiments, an oxide, such as a native oxide, a thermal oxide, or the like is formed at the back side **20B** of the power gating device **20**, such as on the semiconductor substrate **22**, and is used for the bonding. The bonding strength is then improved in a subsequent annealing step, in which the dielectric layer **18** and semiconductor substrate **22** are annealed at a high temperature. After the annealing, bonds, such as fusions bonds, are formed bonding the dielectric layer **18** and semiconductor substrate **22**. For example, the bonds can be covalent bonds between the material of the dielectric layer **18** and the material of the semiconductor substrate **22**. The die connectors **16A** and conductive vias **30** are connected to each other with a one-to-one correspondence. The die connectors **16A** and conductive vias **30** may be in physical contact after the pre-bonding, or may expand to be brought into physical contact during the annealing. Further, during the annealing, the material of the die connectors **16A** and conductive vias **30** (e.g., copper) intermingles, so that metal-to-metal bonds are also formed. Hence, the resulting bonds between the processor device **10** and power gating device **20** are hybrid bonds that include both dielectric-to-dielectric bonds and metal-to-metal bonds.

(52) FIGS. **8**, **9**, **10**, **11**, **12**, **13**, **14**, **15**, and **16** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package **100**, in accordance with some other embodiments. In this embodiment, the integrated circuit package **100** is singulated and included in a package component. Packaging of devices in one package region **302A** is illustrated, but it should be appreciated that any number of package regions may be simultaneously formed. The package region **302A** will be singulated in subsequent processing. The singulated package component may be a fan-out package, such as an integrated fan-out (InFO) package. The fan-out package is then mounted to a package substrate.

(53) In FIG. **8**, a carrier substrate **302** is provided, and a release layer **304** is formed on the carrier substrate **302**. The carrier substrate **302** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier substrate **302** may be a wafer, such that multiple packages can be formed on the carrier substrate **302** simultaneously. The release layer **304** may be formed of a polymer-based material, which may be removed along with the carrier substrate **302** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **304** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In other embodiments, the release layer **304** may be an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer **304** may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier substrate **302**, or may be the like. The top surface of the release layer **304** may be leveled and may have a high degree of planarity.

(54) In FIG. **9**, a redistribution structure **306** may be formed on the release layer **304**. The redistribution structure **306** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **306**

includes dielectric layers **308** and metallization patterns **310** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **308** and metallization patterns **310** than illustrated may be formed in the redistribution structure **306**. The redistribution structure **306** is optional. In some embodiments, a dielectric layer without metallization patterns is formed on the release layer **304** in lieu of the redistribution structure **306**.

(55) In FIG. **10**, conductive vias **312** are formed extending through the topmost dielectric layer **308** of the redistribution structure **306**. Thus, the conductive vias **312** are connected to the metallization patterns **310** of the redistribution structure **306**. The conductive vias **312** are optional, and may be omitted. For example, the conductive vias **312** may (or may not) be omitted in embodiments where the redistribution structure **306** is omitted.

(56) As an example to form the conductive vias **312**, openings can be formed in the topmost dielectric layer **308** of the redistribution structure **306**. A seed layer is then formed over the redistribution structure **306**, e.g., on the topmost dielectric layer **308** and portions of the metallization pattern **310** exposed by the openings in the topmost dielectric layer **308**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In a particular embodiment, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to conductive vias. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the conductive vias **312**.

(57) In FIG. **11**, a singulated integrated circuit package **100** is placed on the redistribution structure **306** (e.g., the topmost dielectric layer **308**). To form the singulated integrated circuit package **100**, an intermediate structure similar to that described with respect to FIG. **3** is obtained. A singulation process is performed by sawing along scribe line regions, e.g., around the device region **102A** (see FIG. **3**). The singulation process includes sawing the dielectric layer **104** and wafer **102**. The singulation process separates the device region **102A** (comprising the processor device **10**) from adjacent device regions (not illustrated) of the wafer **102** to form an integrated circuit package **100** comprising the processor device **10**. As noted above, the power gating device **20** is directly bonded to the processor device **10** in a face-to-face manner, without the use of solder. The resulting integrated circuit package **100** is thus free from solder. After singulation, the dielectric layer **104** and processor device **10** are laterally coterminous.

(58) In FIG. **12**, an encapsulant **314** is formed on and around the conductive vias **312** and integrated circuit package **100**. After formation, the encapsulant **314** encapsulates the conductive vias **312** and integrated circuit package **100**. The encapsulant **314** may be a molding compound, epoxy, or the like. The encapsulant **314** may be applied by compression molding, transfer molding, or the like, and may be formed over the carrier substrate **302** such that the integrated circuit package **100** and/or the conductive vias **312** are buried or covered. The encapsulant **314** may be applied in liquid or semi-liquid form and then subsequently cured. In some embodiments, the encapsulant **314** is different from and comprises a different material than the dielectric layer **104**. A planarization process can then be performed on the encapsulant **314** to expose the conductive vias **312** and the integrated circuit package **100**. The planarization process may remove material of the

conductive vias **312**, encapsulant **314**, dielectric layer **104**, conductive vias **106**, semiconductor substrate **22**, and/or conductive vias **30** until the conductive vias **30**, **106**, **312** are exposed. Top surfaces of the planarized components are coplanar after the planarization process. The planarization process may be, for example, a chemical-mechanical polish (CMP), a grinding process, an etch-back, or the like. In some embodiments, the planarization may be omitted, for example, if the conductive vias **30**, **106**, **312** are already exposed.

(59) In FIG. **13**, a redistribution structure **316** is formed on the encapsulant **314**, conductive vias **312**, and integrated circuit package **100**. The redistribution structure **316** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **316** includes dielectric layers **318** and metallization patterns **320** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **318** and metallization patterns **320** than illustrated may be formed in the redistribution structure **316**. The metallization patterns **320** of the redistribution structure **316** are connected to the conductive vias **30**, **106**, **312**. The metallization patterns **320** of the redistribution structure **316** include power supply source (V.sub.DD) lines, power supply ground (V.sub.SS) lines, and data signal lines, which are electrically coupled to the processor device **10** and power gating device **20** in a similar manner as that described with respect to FIG. **4**.

(60) In FIG. **14**, conductive connectors **322** are formed connected to the metallization patterns **320** of the redistribution structure **316**. The conductive connectors **322** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**. For example, the conductive connectors **322** can be formed on UBMs of the redistribution structure **316**.

(61) In FIG. **15**, a carrier substrate debonding is performed to detach (de-bond) the carrier substrate **302** from the redistribution structure **306**, e.g., the bottommost dielectric layer **308**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer **304** so that the release layer **304** decomposes under the heat of the light and the carrier substrate **302** can be removed. The structure can then be flipped over and placed on, e.g., a tape.

(62) Further, conductive connectors **324** are formed through the bottommost dielectric layer **308** of the redistribution structure **306**. Openings can be formed through the bottommost dielectric layer **308** of the redistribution structure **306**, exposing portions of the metallization patterns **310**. The openings may be formed, for example, using laser drilling, etching, or the like. The conductive connectors **324** are formed in the openings, and are connected to exposed portions of the metallization patterns **310**. The conductive connectors **324** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**.

(63) In FIG. **16**, a singulation process is performed by sawing along scribe line regions, e.g., around the package region **302A**. The singulation process includes sawing the redistribution structures **306**, **316** and encapsulant **314**. The singulation process separates the package region **302A** from adjacent package regions (not illustrated) to form an integrated circuit package **300**. After singulation, the redistribution structures **306**, **316** (e.g., the dielectric layers **308**, **318**) and the encapsulant **314** are laterally coterminous.

(64) Another integrated circuit package **400** can be attached to the integrated circuit package **300** to form a package-on-package structure. The integrated circuit package **400** may be a memory package. The integrated circuit package **400** can be attached to the integrated circuit package **300** before or after the integrated circuit package **300** is singulated. The integrated circuit package **400** includes a substrate **402** and one or more dies **404** connected to the substrate **402**. In some embodiments (not shown) one or more stacks of dies **404** are connected to the substrate **402**. The substrate **402** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. In some embodiments, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic

phosphide, gallium indium phosphide, combinations of these, and the like, may also be used. Additionally, the substrate **402** may be a silicon-on-insulator (SOI) substrate. Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, SOI, silicon germanium on insulator (SGOI), or combinations thereof. The substrate **402** is, in one alternative embodiment, based on an insulating core such as a fiberglass reinforced resin core. One example core material is fiberglass resin such as FR4. Alternatives for the core material include bismaleimide-triazine (BT) resin, or alternatively, other printed circuit board (PCB) materials or films. Build up films such as Ajinomoto build-up film (ABF) or other laminates may be used for substrate **402**.

(65) The substrate **402** may include active and passive devices (not shown). As one of ordinary skill in the art will recognize, a wide variety of devices such as transistors, capacitors, resistors, combinations of these, and the like may be used to generate the structural and functional requirements of the design for the integrated circuit package **400**. The devices may be formed using any suitable methods. The substrate **402** may also include metallization layers (not shown) and through vias. The metallization layers may be formed over the active and passive devices and are designed to connect the various devices to form functional circuitry. The metallization layers may be formed of alternating layers of dielectric (e.g., low-k dielectric material) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material and may be formed through any suitable process (such as deposition, damascene, dual damascene, or the like). In some embodiments, the substrate **402** is substantially free of active and passive devices.

(66) The substrate **402** may have bond pads **406** on a side of the substrate **402**, to connect to the conductive connectors **324**. In some embodiments, the bond pads **406** are formed by forming recesses (not shown) into dielectric layers (not shown) on the side of the substrate **402**. The recesses may be formed to allow the bond pads **406** to be embedded into the dielectric layers. In other embodiments, the recesses are omitted as the bond pads **406** may be formed on the dielectric layer. In some embodiments, the bond pads **406** include a thin seed layer (not shown) made of copper, titanium, nickel, gold, palladium, the like, or a combination thereof. The conductive material of the bond pads **406** may be deposited over the thin seed layer. The conductive material may be formed by an electro-chemical plating process, an electroless plating process, CVD, ALD, PVD, the like, or a combination thereof. In an embodiment, the conductive material of the bond pads **406** is copper, tungsten, aluminum, silver, gold, the like, or a combination thereof.

(67) In an embodiment, the bond pads **406** are UBMs that include three layers of conductive materials, such as a layer of titanium, a layer of copper, and a layer of nickel. For example, the bond pads **406** may be formed from copper, may be formed on a layer of titanium (not shown), and have a nickel finish, which may improve the shelf life of the integrated circuit package **400**, which may be particularly advantageous when the integrated circuit package **400** is a memory device such as a DRAM module. However, one of ordinary skill in the art will recognize that there are many suitable arrangements of materials and layers, such as an arrangement of chrome/chrome-copper alloy/copper/gold, an arrangement of titanium/titanium tungsten/copper, or an arrangement of copper/nickel/gold, that are suitable for the formation of the bond pads **406**. Any suitable materials or layers of material that may be used for the bond pads **406** are fully intended to be included within the scope of the current application.

(68) In the illustrated embodiment, the dies **404** are connected to the substrate **402** by conductive bumps, although other connections may be used, such as wire bonds. In an embodiment, the dies **404** are stacked memory dies. For example, the dies **404** may be memory dies such as low-power (LP) double data rate (DDR) memory modules, such as LPDDR1, LPDDR2, LPDDR3, LPDDR4, or the like.

(69) The dies **404** and the wire bonds (when present) may be encapsulated by a molding material **410**. The molding material **410** may be molded on the dies **404** and the wire bonds, for example, using compression molding. In some embodiments, the molding material **410** is a molding

compound, a polymer, an epoxy, silicon oxide filler material, the like, or a combination thereof. A curing process may be performed to cure the molding material **410**; the curing process may be a thermal curing, a UV curing, the like, or a combination thereof. In some embodiments, the dies **404** are buried in the molding material **410**, and after the curing of the molding material **410**, a planarization step, such as a grinding, is performed to remove excess portions of the molding material **410** and provide a substantially planar surface for the integrated circuit package **400**. (70) After the integrated circuit package **400** is formed, the integrated circuit package **400** is attached to the integrated circuit package **300** by way of the conductive connectors **324**. The conductive connectors **324** can be connected to the bond pads **406** by reflowing the conductive connectors **324**. The dies **404** may thus be electrically coupled to the integrated circuit package **100** through the conductive connectors **324**, the conductive vias **312**, and the redistribution structures **306**, **316**.

(71) In some embodiments, a solder resist (not shown) is formed on the side of the substrate **402** opposing the dies **404**. The conductive connectors **324** may be disposed in openings in the solder resist to be connected to conductive features (e.g., the bond pads **406**) in the substrate **402**. The solder resist may be used to protect areas of the substrate **402** from external damage.

(72) In some embodiments, the conductive connectors **324** have an epoxy flux (not shown) formed thereon before they are reflowed with at least some of the epoxy portion of the epoxy flux remaining after the integrated circuit package **400** is attached to the redistribution structure **306**.

(73) In some embodiments, an underfill (not shown) is formed between the redistribution structure **306** and the substrate **402**, and surrounding the conductive connectors **324**. The underfill may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **324**. The underfill may be formed by a capillary flow process after the integrated circuit package **400** is attached or may be formed by a suitable deposition method before the integrated circuit package **400** is attached. In embodiments where the epoxy flux is formed, it may act as the underfill.

(74) The package-on-package structure is then flipped and attached to a package substrate **200** using the conductive connectors **322**. The package substrate **200** may be similar to the package substrate **200** described with respect to FIG. 6. For example, the package substrate **200** can include bond pads **202**, which are connected to the conductive connectors **322**.

(75) FIG. 17 is a cross-sectional view of an integrated circuit package **100** and a system implementing the integrated circuit package **100**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. 16, except the processor device **10** and power gating device **20** are directly bonded in a face-to-back manner, similar to the hybrid bonding described with respect to FIG. 7. The metallization patterns **320** of the redistribution structure **316** (e.g., V.sub.DD and V.sub.SS lines) are thus connected to the die connectors **26**, and are electrically coupled to the processor device **10** by the conductive vias **30**.

(76) FIG. 18 is a cross-sectional view of an integrated circuit package **100** and a system implementing the integrated circuit package **100**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. 16, except the redistribution structure **306**, the conductive vias **312**, and the integrated circuit package **400** are omitted.

(77) FIG. 19 is a cross-sectional view of an integrated circuit package **100** and a system implementing the integrated circuit package **100**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. 17, except the redistribution structure **306**, the conductive vias **312**, and the integrated circuit package **400** are omitted.

(78) FIGS. 20, 21, 22, 23, and 24 are cross-sectional views of intermediate steps during a process for forming an integrated circuit package **500**, in accordance with some embodiments. The integrated circuit package **500** is formed by stacking multiple levels of semiconductor device, each encapsulated in an encapsulating material. Packaging of devices in one package region **502A** is

illustrated, but it should be appreciated that any number of package regions may be simultaneously formed. The package region **502A** will be singulated in subsequent processing. The singulated package component may be a multi-stacked (MUST) package. The MUST package is then mounted to a package substrate.

(79) In FIG. **20**, a carrier substrate **502** is provided, and a release layer **504** is formed on the carrier substrate **502**. The carrier substrate **502** can be formed in a similar manner and of similar materials as the carrier substrate **302** described with respect to FIG. **8**. The release layer **504** can be formed in a similar manner and of similar materials as the release layer **304** described with respect to FIG. **8**. A processor device **10** similar to that described with respect to FIG. **1** (e.g., comprising a semiconductor substrate **12**, interconnect structure **14**, die connectors **16**, and dielectric layer **18**) is then placed on the release layer **504**.

(80) In FIG. **21**, an encapsulant **506** is formed on and around the processor device **10**. The encapsulant **506** can be formed in a similar manner and of similar materials as the encapsulant **314** described with respect to FIG. **12**. A planarization process can be performed on the encapsulant **506**, if needed, to expose the die connectors **16**.

(81) In FIG. **22**, a redistribution structure **510** is formed on the encapsulant **506** and processor device **10**. The redistribution structure **510** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **510** includes dielectric layers **512** and metallization patterns **514** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **512** and metallization patterns **514** than illustrated may be formed in the redistribution structure **510**. The metallization patterns **514** of the redistribution structure **510** are connected to the die connectors **16** of the processor device **10**.

(82) In FIG. **23**, conductive vias **516** are formed extending through the topmost dielectric layer **512** of the redistribution structure **510**. The conductive vias **516** can be formed in a similar manner and of similar materials as the conductive vias **312** described with respect to FIG. **10**.

(83) One or more power gating devices **20** are placed on the redistribution structure **510** (e.g., the topmost dielectric layer **512**). Each power gating device **20** can be similar to that described with respect to FIG. **1** (e.g., comprising a semiconductor substrate **22**, an interconnect structure **24**, die connectors **26**, and a dielectric layer **28**). The power gating devices **20** will be electrically coupled to the processor device **10** through the redistribution structures of the MUST package, and so the formation of the conductive vias **30** (see FIG. **1**) is unnecessary. Manufacturing costs of the power gating devices **20** may thus be reduced.

(84) In FIG. **24**, an encapsulant **518** is formed on and around the power gating devices **20** and conductive vias **516**. The encapsulant **518** can be formed in a similar manner and of similar materials as the encapsulant **314** described with respect to FIG. **12**. A planarization process can be performed on the encapsulant **518**, if needed, to expose the conductive vias **516** and die connectors **26**.

(85) FIGS. **25**, **26**, **27**, and **28** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package **500**, in accordance with some embodiments. In this embodiment, the integrated circuit package **500** is further processed to include a redistribution structure, and is directly mounted to a package substrate.

(86) In FIG. **25**, a redistribution structure **520** is formed on the encapsulant **518** and power gating devices **20**. The redistribution structure **520** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **520** includes dielectric layers **522** and metallization patterns **524** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **522** and metallization patterns **524** than illustrated may be formed in the redistribution structure **520**. The metallization patterns **524** of the redistribution structure **520** are connected to the conductive vias **516** and the die connectors **26** of the power gating devices **20**. Thus, the processor device **10** is electrically coupled

to the metallization patterns **524** of the redistribution structure **520** through the conductive vias **516** and the metallization patterns **514** of the redistribution structure **510**.

(87) The metallization patterns **524** of the redistribution structure **520** include power supply source (V.sub.DD) lines, power supply ground (V.sub.SS) lines, and data signal lines, which are electrically coupled to input terminals of the power gating devices **20**. The metallization patterns **524** of the redistribution structure **520** also include lines that electrically couple the conductive vias **516** to output terminals of the power gating devices **20**. Thus, a first subset of the circuit blocks of the processor device **10** are electrically coupled to the V.sub.DD and V.sub.SS lines through the power gating device **20** and conductive vias **516**. The first subset of the circuit blocks can thus be turned on and off by the power gating device **20**. A second subset of the circuit blocks of the processor device **10** are electrically coupled to the V.sub.DD and V.sub.SS lines through the conductive vias **516**, bypassing the power gating device **20**. The second subset of the circuit blocks are thus permanently electrically coupled to power and ground. The second subset of the circuit blocks may be essential circuit blocks that are always turned on. The data signal lines are electrically coupled to the processor device **10** and power gating device **20**. The processor device **10** may send control signals to the power gating device **20** over the data signal lines, and may receive power/ground signals from the power gating device **20** over the data signal lines.

(88) In FIG. **26**, conductive connectors **526** are formed connected to the metallization patterns **524** of the redistribution structure **520**. The conductive connectors **526** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**.

(89) In FIG. **27**, a carrier substrate debonding is performed to detach (de-bond) the carrier substrate **502** from the redistribution structure **510**, e.g., the bottommost dielectric layer **512**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer **504** so that the release layer **504** decomposes under the heat of the light and the carrier substrate **502** can be removed. The structure can then be flipped over and placed on, e.g., a tape.

(90) In FIG. **28**, a singulation process is performed by sawing along scribe line regions, e.g., around the package region **502A**. The singulation process includes sawing the redistribution structures **510**, **520** and encapsulants **506**, **518**. The singulation process separates the package region **502A** from adjacent package regions (not illustrated) to form an integrated circuit package **500**. After singulation, the redistribution structures **510**, **520** (e.g., the dielectric layers **512**, **522**) and the encapsulants **506**, **518** are laterally coterminous.

(91) The integrated circuit package **500** is then attached to a package substrate **200** using the conductive connectors **526**. The package substrate **200** may be similar to the package substrate **200** described with respect to FIG. **6**. For example, the package substrate **200** can include bond pads **202**, which are connected to the conductive connectors **526**.

(92) FIGS. **29**, **30**, **31**, **32**, **33**, and **34** are cross-sectional views of intermediate steps during a process for forming a system implementing the integrated circuit package **500**, in accordance with some embodiments. In this embodiment, the integrated circuit package **500** is singulated and included in a package component. Packaging of devices in one package region **602A** is illustrated, but it should be appreciated that any number of package regions may be simultaneously formed. The package region **602A** will be singulated in subsequent processing. The singulated package component may be a fan-out package, such as an integrated fan-out (InFO) package. The fan-out package is then mounted to a package substrate.

(93) In FIG. **29**, a carrier substrate **602** is provided, and a release layer **604** is formed on the carrier substrate **602**. The carrier substrate **602** can be formed in a similar manner and of similar materials as the carrier substrate **302** described with respect to FIG. **8**. The release layer **604** can be formed in a similar manner and of similar materials as the release layer **304** described with respect to FIG. **8**.

(94) A redistribution structure **606** may be formed on the release layer **604**. The redistribution structure **606** can be formed in a similar manner and of similar materials as the redistribution

structure **108** described with respect to FIG. **4**. The redistribution structure **606** includes dielectric layers **608** and metallization patterns **610** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **608** and metallization patterns **610** than illustrated may be formed in the redistribution structure **606**. The redistribution structure **606** is optional. In some embodiments, a dielectric layer without metallization patterns is formed on the release layer **604** in lieu of the redistribution structure **606**.

(95) In FIG. **30**, conductive vias **612** are formed extending through the topmost dielectric layer **608** of the redistribution structure **606**. Thus, the conductive vias **612** are connected to the metallization patterns **610** of the redistribution structure **606**. The conductive vias **612** can be formed in a similar manner and of similar materials as the conductive vias **312** described with respect to FIG. **10**. The conductive vias **612** are optional, and may be omitted. For example, the conductive vias **612** may (or may not) be omitted in embodiments where the redistribution structure **606** is omitted.

(96) A singulated integrated circuit package **500** is placed on the redistribution structure **606** (e.g., the topmost dielectric layer **608**), adjacent the conductive vias **612**. To form the singulated integrated circuit package **500**, an intermediate structure similar to that described with respect to FIG. **24** is obtained. A carrier substrate debonding is performed to detach (de-bond) the intermediate structure from the carrier substrate **502** (see FIG. **24**). A singulation process is then performed by sawing along scribe line regions, e.g., around the package region **502A** (see FIG. **24**). The singulation process includes sawing the redistribution structure **510** and encapsulants **506**, **518**. The singulation process separates the package region **502A** from adjacent package regions (not illustrated) to form an integrated circuit package **500**. After singulation, the redistribution structure **510** (e.g., the dielectric layers **512**) and the encapsulants **506**, **518** are laterally coterminous.

(97) In FIG. **31**, an encapsulant **614** is formed on and around the conductive vias **612** and integrated circuit package **500**. The encapsulant **614** can be formed in a similar manner and of similar materials as the encapsulant **314** described with respect to FIG. **12**. In some embodiments, the encapsulant **614** is different from and comprises a different material than the encapsulants **506**, **518**. A planarization process can be performed on the encapsulant **614**, if needed, to expose the die connectors **26** and conductive vias **516**, **612**.

(98) In FIG. **32**, a redistribution structure **616** is formed on the encapsulant **614**, conductive vias **612**, and integrated circuit package **500**. The redistribution structure **616** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **616** includes dielectric layers **618** and metallization patterns **620** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **618** and metallization patterns **620** than illustrated may be formed in the redistribution structure **616**. The metallization patterns **620** of the redistribution structure **616** are connected to the die connectors **26** and conductive vias **516**, **612**. The metallization patterns **620** of the redistribution structure **616** include power supply source (V.sub.DD) lines, power supply ground (V.sub.SS) lines, and data signal lines, which are electrically coupled to the processor device **10** and power gating device **20** in a similar manner as that described with respect to FIG. **4**.

(99) Conductive connectors **622** are formed connected to the metallization patterns **620** of the redistribution structure **616**. The conductive connectors **622** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**. For example, the conductive connectors **622** can be formed on UBMs of the redistribution structure **616**.

(100) In FIG. **33**, a carrier substrate debonding is performed to detach (de-bond) the carrier substrate **602** from the redistribution structure **606**, e.g., the bottommost dielectric layer **608**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer **604** so that the release layer **604** decomposes under the heat of the light and the carrier substrate **602** can be removed. The structure can then be flipped over and placed on, e.g., a tape.

(101) Further, conductive connectors **624** are formed through the bottommost dielectric layer **608** of the redistribution structure **606**. Openings can be formed through the bottommost dielectric layer **608** of the redistribution structure **606**, exposing portions of the metallization patterns **610**. The openings may be formed, for example, using laser drilling, etching, or the like. The conductive connectors **624** are formed in the openings, and are connected to exposed portions of the metallization patterns **610**. The conductive connectors **624** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. 5.

(102) In FIG. 34, a singulation process is performed by sawing along scribe line regions, e.g., around the package region **602A**. The singulation process includes sawing the redistribution structures **606**, **616** and encapsulant **614**. The singulation process separates the package region **602A** from adjacent package regions (not illustrated) to form an integrated circuit package **600**. After singulation, the redistribution structures **606**, **616** (e.g., the dielectric layers **608**, **618**) and the encapsulant **614** are laterally coterminous.

(103) Another integrated circuit package **400** can be attached to the integrated circuit package **600** to form a package-on-package structure. The integrated circuit package **400** can be attached to the integrated circuit package **600** before or after the integrated circuit package **600** is singulated. The integrated circuit package **400** may be similar to the integrated circuit package **400** described with respect to FIG. 16. For example, the integrated circuit package **400** can include bond pads **406**, which are connected to the conductive connectors **624**.

(104) The package-on-package structure is then flipped and attached to a package substrate **200** using the conductive connectors **622**. The package substrate **200** may be similar to the package substrate **200** described with respect to FIG. 6. For example, the package substrate **200** can include bond pads **202**, which are connected to the conductive connectors **622**.

(105) FIGS. 35, 36, 37, 38, 39, and 40 are cross-sectional views of intermediate steps during a process for forming an integrated circuit package **700**, in accordance with some other embodiments. The integrated circuit package **700** is formed by singulating processor devices **10** and power gating devices **20**, and including them in a package component. Packaging of devices in one package region **702A** is illustrated, but it should be appreciated that any number of package regions may be simultaneously formed. The package region **702A** will be singulated in subsequent processing. The singulated package component may be a fan-out package, such as an integrated fan-out (InFO) package. The fan-out package is then mounted to a package substrate.

(106) In FIG. 35, a carrier substrate **702** is provided, and a release layer **704** is formed on the carrier substrate **702**. The carrier substrate **702** can be formed in a similar manner and of similar materials as the carrier substrate **302** described with respect to FIG. 8. The release layer **704** can be formed in a similar manner and of similar materials as the release layer **304** described with respect to FIG. 8. Conductive vias **706** are then formed on and extending away from the release layer **704**. The conductive vias **706** can be formed in a similar manner and of similar materials as the conductive vias **312** described with respect to FIG. 10.

(107) A processor device **10** similar to that described with respect to FIG. 1 (e.g., comprising a semiconductor substrate **12**, interconnect structure **14**, die connectors **16**, and dielectric layer **18**) is then placed on the release layer **704**, adjacent the conductive vias **706**. In this embodiment, the processor device **10** further includes conductive vias **708**, which are formed extending through the semiconductor substrate **12**. The conductive vias **708** are electrically coupled to metallization patterns of the interconnect structure **14**.

(108) In FIG. 36, an encapsulant **710** is formed on and around the conductive vias **706** and processor device **10**. The encapsulant **710** can be formed in a similar manner and of similar materials as the encapsulant **314** described with respect to FIG. 12. A planarization process can be performed on the encapsulant **710**, if needed, to expose the die connectors **16** and conductive vias **706**.

(109) In FIG. 37, a redistribution structure **712** is formed on the encapsulant **710**, conductive vias

706, and processor device **10**. The redistribution structure **712** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **712** includes dielectric layers **714** and metallization patterns **716** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **714** and metallization patterns **716** than illustrated may be formed in the redistribution structure **712**. The metallization patterns **716** of the redistribution structure **712** are connected to the die connectors **16** and conductive vias **706**.

(110) Conductive connectors **718** are formed connected to the metallization patterns **716** of the redistribution structure **712**. The conductive connectors **718** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**. For example, the conductive connectors **718** can be formed on UBMs of the redistribution structure **712**.

(111) In FIG. **38**, a carrier substrate debonding is performed to detach (de-bond) the carrier substrate **702** from the processor device **10**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer **704** so that the release layer **704** decomposes under the heat of the light and the carrier substrate **702** can be removed. The structure can then be flipped over and placed on, e.g., a tape.

(112) A redistribution structure **720** is formed on the encapsulant **710**, conductive vias **706**, and processor device **10**. The redistribution structure **720** can be formed in a similar manner and of similar materials as the redistribution structure **108** described with respect to FIG. **4**. The redistribution structure **720** includes dielectric layers **722** and metallization patterns **724** (sometimes referred to as redistribution layers or redistribution lines). More or fewer dielectric layers **722** and metallization patterns **724** than illustrated may be formed in the redistribution structure **720**. The metallization patterns **724** of the redistribution structure **720** are connected to the conductive vias **706**, **708**.

(113) In FIG. **39**, a power gating device **20** similar to that described with respect to FIG. **1** (e.g., comprising a semiconductor substrate **22**, interconnect structure **24**, and die connectors **26**) is attached to the redistribution structure **720** with conductive connectors **726**. The power gating device **20** will be electrically coupled to the processor device **10** through the redistribution structures **712**, **720** (e.g., the metallization patterns **716**, **724**), and so the formation of the conductive vias **30** (see FIG. **1**) is unnecessary. Further, the dielectric layer **28** (see FIG. **1**) can also be omitted from the power gating device **20**. Manufacturing costs of the power gating device **20** may thus be reduced.

(114) In some embodiments, the power gating device **20** is attached to conductive bumps, such as micro bumps. For example, die connectors **728** can be formed extending through the bottommost dielectric layer **722** of the redistribution structure **720**. As an example to form the die connectors **728**, openings can be formed through the bottommost dielectric layer **722** of the redistribution structure **720**, exposing portions of the metallization patterns **724**. The openings may be formed, for example, using laser drilling, etching, or the like. The die connectors **728** can be plated or deposited in the openings, and are connected to exposed portions of the metallization patterns **724**. Conductive connectors **726** are formed connecting respective pairs of the die connectors **26**, **728**. The conductive connectors **726** can be formed in a similar manner and of similar materials as the conductive connectors **114** described with respect to FIG. **5**. In this embodiment, the back side of the processor device **10** faces the front side of the power gating device **20**.

(115) The metallization patterns **716** of the redistribution structure **712** include power supply source (V.sub.DD) lines, power supply ground (V.sub.SS) lines, and data signal lines. A first subset of the V.sub.DD and V.sub.SS lines are connected to the die connectors **16** of the processor device **10**. Thus, a first subset of the circuit blocks of the processor device **10** are electrically connected to the V.sub.DD and V.sub.SS lines of the redistribution structure **712**, bypassing the power gating device **20**. The first subset of the circuit blocks are thus permanently electrically coupled to power

and ground. The first subset of the circuit blocks may be essential circuit blocks that are always turned on. A second subset of the V.sub.DD and V.sub.SS lines are electrically coupled to input terminals of the power gating device **20** through the conductive vias **706** and the metallization patterns **724** of the redistribution structure **720**. The metallization patterns **724** of the redistribution structure **720** also include lines that electrically couple the processor device **10** to output terminals of the power gating device **20**. Thus, a second subset of the circuit blocks of the processor device **10** are electrically coupled to the V.sub.DD and V.sub.SS lines through the power gating device **20**, redistribution structure **720**, and conductive vias **706**, **708**. The second subset of the circuit blocks can thus be turned on and off by the power gating device **20**. In some embodiments, some of the conductive vias **708** are connected to data signal lines of the redistribution structure **720**. The data signal lines are electrically coupled to the processor device **10** and power gating device **20**. The processor device **10** may send control signals to the power gating device **20** over the data signal lines, and may receive power/ground signals from the power gating device **20** over the data signal lines.

(116) An underfill **730** is then formed between the power gating device **20** and the redistribution structure **720**, surrounding the conductive connectors **726**. The underfill **730** may be formed by a capillary flow process after the power gating device **20** is attached or may be formed by a suitable deposition method before the power gating device **20** is attached.

(117) An encapsulant **732** is formed on and around the power gating device **20** and the underfill **730**. The encapsulant **732** can be formed in a similar manner and of similar materials as the encapsulant **314** described with respect to FIG. **12**. A planarization process can be performed on the encapsulant **710**, if needed, to expose the power gating device **20**.

(118) In FIG. **40**, a singulation process is performed by sawing along scribe line regions, e.g., around the package region **702A**. The singulation process includes sawing the redistribution structures **712**, **720** and encapsulants **710**, **732**. The singulation process separates the package region **702A** from adjacent package regions (not illustrated) to form an integrated circuit package **700**. After singulation, the redistribution structures **712**, **720** (e.g., the dielectric layers **714**, **722**) and the encapsulants **710**, **732** are laterally coterminous.

(119) The integrated circuit package **700** is then attached to a package substrate **200** using the conductive connectors **718**. The package substrate **200** may be similar to the package substrate **200** described with respect to FIG. **6**. For example, the package substrate **200** can include bond pads **202**, which are connected to the conductive connectors **718**.

(120) FIG. **41** is a cross-sectional view of an integrated circuit package **700**, in accordance with some other embodiments. This embodiment is similar to the embodiment described with respect to FIG. **40**, except the front side of the processor device **10** faces the front side of the power gating device **20**. The metallization patterns **724** of the redistribution structure **720** (e.g., V.sub.DD and V.sub.SS lines) are thus directly connected to the die connectors **16**. The conductive vias **706**, **708** are thus used for data signal lines.

(121) Embodiments may achieve advantages. Forming integrated circuit packages with power gating devices **20** that are separate from the processor devices **10** allows the power gating device **20** to be formed with larger active devices than the processor device **10**. Switch transistors of a large technology node may thus be used for power delivery. Switch transistors of a large technology node can accommodate greater rush currents and have a smaller voltage drop (e.g., IR drop) when the circuit blocks of the processor device **10** are turned on, and also have less power leakage. The power consumption of the resulting integrated circuit packages may thus be reduced.

(122) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure.

Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(123) In an embodiment, a device includes: a processor die including circuit blocks, the circuit blocks including active devices of a first technology node; a power gating die including power semiconductor devices of a second technology node, the second technology node larger than the first technology node; and a first redistribution structure including first metallization patterns, the first metallization patterns including power supply source lines and power supply ground lines, where a first subset of the circuit blocks is electrically coupled to the power supply source lines and the power supply ground lines through the power semiconductor devices, and a second subset of the circuit blocks is permanently electrically coupled to the power supply source lines and the power supply ground lines.

(124) In some embodiments, the device further includes: a package substrate; and a conductive connector connected to the package substrate and the first metallization patterns of the first redistribution structure. In some embodiments of the device, the processor die and the power gating die are directly bonded by dielectric-to-dielectric bonds and by metal-to-metal bonds, and the device further includes: a conductive via connected to the processor die and the first metallization patterns of the first redistribution structure; and a dielectric layer surrounding the conductive via and the power gating die, the dielectric layer and the power gating die each disposed between the first redistribution structure and the processor die, the first redistribution structure, the dielectric layer, and the processor die being laterally coterminous. In some embodiments of the device, the processor die and the power gating die are directly bonded by dielectric-to-dielectric bonds and by metal-to-metal bonds, and the device further includes: a conductive via connected to the processor die and the first metallization patterns of the first redistribution structure; a dielectric layer surrounding the conductive via and the power gating die, the dielectric layer and the power gating die each disposed between the first redistribution structure and the processor die, the dielectric layer and the processor die being laterally coterminous; and an encapsulant surrounding the processor die and the dielectric layer, the encapsulant and the first redistribution structure being laterally coterminous. In some embodiments of the device, the first metallization patterns of the first redistribution structure are connected to the power gating die, and the device further includes: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution structure including second metallization patterns, the processor die connected to the second metallization patterns; a first encapsulant surrounding the power gating die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; and a second encapsulant surrounding the processor die, the second encapsulant, the second redistribution structure, the first encapsulant, and the first redistribution structure being laterally coterminous. In some embodiments of the device, the first metallization patterns of the first redistribution structure are connected to the power gating die, and the device further includes: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution structure including second metallization patterns, the processor die connected to the second metallization patterns; a first encapsulant surrounding the power gating die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; a second encapsulant surrounding the processor die, the second encapsulant, the second redistribution structure, and the first encapsulant being laterally coterminous; and a third encapsulant surrounding the second encapsulant, the second redistribution structure, and the first encapsulant, the third encapsulant and the first redistribution structure being laterally coterminous. In some embodiments of the device, the first metallization patterns of the first redistribution structure are connected to the processor die,

and the device further includes: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution structure including second metallization patterns, the second metallization patterns connected to the processor die and the power gating die; a first encapsulant surrounding the processor die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; and a second encapsulant surrounding the power gating die, the second encapsulant, the first encapsulant, the second redistribution structure, and the first redistribution structure being laterally coterminous. In some embodiments of the device, the power gating die is operable at runtime to receive a control signal from the processor die, and to turn one of the first subset of the circuit blocks of the processor die on or off responsive to the control signal. In some embodiments of the device, the active devices of the processor die are complementary metal-oxide-semiconductor devices, and the power semiconductor devices of the power gating die are insulated-gate bipolar transistor devices or double diffused metal-oxide-semiconductor devices.

(125) In an embodiment, a device includes: a processor die including circuit blocks, the circuit blocks including active devices of a first technology node; a power gating die directly bonded to the processor die by dielectric-to-dielectric bonds and by metal-to-metal bonds, the power gating die including power semiconductor devices of a second technology node, the second technology node larger than the first technology node, the power semiconductor devices electrically coupled to a first subset of the circuit blocks; a dielectric layer surrounding the power gating die; first conductive vias extending through the dielectric layer, the first conductive vias electrically coupled to a second subset of the circuit blocks; and a first redistribution structure including first metallization patterns, the first metallization patterns including power supply source lines connected to the first conductive vias and the power gating die; a package substrate; and first conductive connectors connecting the package substrate to the first metallization patterns of the first redistribution structure.

(126) In some embodiments of the device, the processor die, the dielectric layer, and the first redistribution structure are laterally coterminous. In some embodiments of the device, the processor die and the dielectric layer are laterally coterminous, and the device further includes: an encapsulant surrounding the processor die and the dielectric layer, the encapsulant and the first redistribution structure being laterally coterminous. In some embodiments, the device further includes: a second redistribution structure including second metallization patterns, the encapsulant disposed between the first redistribution structure and the second redistribution structure; second conductive vias extending through the encapsulant, the second conductive vias connected to the second metallization patterns of the second redistribution structure and the first metallization patterns of the first redistribution structure; a memory package; and second conductive connectors connecting the memory package to the second metallization patterns of the second redistribution structure. In some embodiments of the device, the active devices of the processor die are complementary metal-oxide-semiconductor devices, and the power semiconductor devices of the power gating die are insulated-gate bipolar transistor devices or double diffused metal-oxide-semiconductor devices. In some embodiments of the device, the processor die and the power gating die are directly bonded in a face-to-face manner. In some embodiments of the device, the processor die and the power gating die are directly bonded in a face-to-back manner.

(127) In an embodiment, a method includes: bonding a power gating die to a wafer, the wafer including a processor die, the power gating die and the processor die including active devices of different pitches, the power gating die including first conductive vias; forming a dielectric layer around the power gating die and on the wafer; patterning openings in first dielectric layer, the openings exposing die connectors of the processor die; plating a conductive material in the openings and on the die connectors; planarizing the conductive material to form second conductive vias in the openings, the planarizing exposing the first conductive vias of the power gating die; and

sawing the dielectric layer and the wafer to singulate the processor die.

(128) In some embodiments, the method further includes: before the sawing, forming a redistribution structure on the power gating die, the dielectric layer, and the second conductive vias, the sawing including sawing the redistribution structure. In some embodiments, the method further includes, after the sawing: encapsulating the power gating die and the dielectric layer with an encapsulant; and forming a redistribution structure on the encapsulant, the power gating die, the dielectric layer, and the second conductive vias. In some embodiments of the method, the power gating die includes active devices of a larger technology node than the processor die.

(129) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a processor die comprising circuit blocks, the circuit blocks comprising active devices of a first technology node; a power gating die comprising power semiconductor devices of a second technology node, a pitch between adjacent ones of the power semiconductor devices of the second technology node being larger than a pitch between adjacent ones of the active devices of the first technology node; and a first redistribution structure comprising first metallization patterns, the first metallization patterns comprising power supply source lines and power supply ground lines, wherein a first subset of the circuit blocks is electrically coupled to the power supply source lines and the power supply ground lines through the power semiconductor devices, and a second subset of the circuit blocks is permanently electrically coupled to the power supply source lines and the power supply ground lines.

2. The device of claim 1 further comprising: a package substrate; and a conductive connector connected to the package substrate and the first metallization patterns of the first redistribution structure.

3. The device of claim 1, wherein the processor die and the power gating die are directly bonded by dielectric-to-dielectric bonds and by metal-to-metal bonds, and the device further comprises: a conductive via connected to the processor die and the first metallization patterns of the first redistribution structure; and a dielectric layer surrounding the conductive via and the power gating die, the dielectric layer and the power gating die each disposed between the first redistribution structure and the processor die, the first redistribution structure, the dielectric layer, and the processor die being laterally coterminous.

4. The device of claim 1, wherein the processor die and the power gating die are directly bonded by dielectric-to-dielectric bonds and by metal-to-metal bonds, and the device further comprises: a conductive via connected to the processor die and the first metallization patterns of the first redistribution structure; a dielectric layer surrounding the conductive via and the power gating die, the dielectric layer and the power gating die each disposed between the first redistribution structure and the processor die, the dielectric layer and the processor die being laterally coterminous; and an encapsulant surrounding the processor die and the dielectric layer, the encapsulant and the first redistribution structure being laterally coterminous.

5. The device of claim 1, wherein the first metallization patterns of the first redistribution structure are connected to the power gating die, and the device further comprises: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution

structure comprising second metallization patterns, the processor die connected to the second metallization patterns; a first encapsulant surrounding the power gating die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; and a second encapsulant surrounding the processor die, the second encapsulant, the second redistribution structure, the first encapsulant, and the first redistribution structure being laterally coterminous.

6. The device of claim 1, wherein the first metallization patterns of the first redistribution structure are connected to the power gating die, and the device further comprises: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution structure comprising second metallization patterns, the processor die connected to the second metallization patterns; a first encapsulant surrounding the power gating die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; a second encapsulant surrounding the processor die, the second encapsulant, the second redistribution structure, and the first encapsulant being laterally coterminous; and a third encapsulant surrounding the second encapsulant, the second redistribution structure, and the first encapsulant, the third encapsulant and the first redistribution structure being laterally coterminous.

7. The device of claim 1, wherein the first metallization patterns of the first redistribution structure are connected to the processor die, and the device further comprises: a second redistribution structure disposed between the processor die and the power gating die, the second redistribution structure comprising second metallization patterns, the second metallization patterns connected to the processor die and the power gating die; a first encapsulant surrounding the processor die; a conductive via extending through the first encapsulant, the conductive via connected to the first metallization patterns of the first redistribution structure and the second metallization patterns of the second redistribution structure; and a second encapsulant surrounding the power gating die, the second encapsulant, the first encapsulant, the second redistribution structure, and the first redistribution structure being laterally coterminous.

8. The device of claim 1, wherein the power gating die comprises an integrated circuit configured to receive a control signal from the processor die, and to turn one of the first subset of the circuit blocks of the processor die on or off responsive to the control signal.

9. The device of claim 1, wherein the active devices of the processor die are complementary metal-oxide-semiconductor devices, and the power semiconductor devices of the power gating die are insulated-gate bipolar transistor devices or double diffused metal-oxide-semiconductor devices.

10. A device comprising: a processor die comprising circuit blocks, the circuit blocks comprising active devices of a first technology node; a power gating die directly bonded to the processor die by dielectric-to-dielectric bonds and by metal-to-metal bonds, the power gating die comprising power semiconductor devices of a second technology node, the second technology node larger than the first technology node, the power semiconductor devices electrically coupled to a first subset of the circuit blocks; a dielectric layer surrounding the power gating die; first conductive vias extending through the dielectric layer, the first conductive vias electrically coupled to a second subset of the circuit blocks; and a first redistribution structure comprising first metallization patterns, the first metallization patterns comprising power supply source lines connected to the first conductive vias and the power gating die; a package substrate; and first conductive connectors connecting the package substrate to the first metallization patterns of the first redistribution structure.

11. The device of claim 10, wherein the processor die, the dielectric layer, and the first redistribution structure are laterally coterminous.

12. The device of claim 10, wherein the processor die and the dielectric layer are laterally coterminous, and the device further comprises: an encapsulant surrounding the processor die and

the dielectric layer, the encapsulant and the first redistribution structure being laterally coterminous.

13. The device of claim 12 further comprising: a second redistribution structure comprising second metallization patterns, the encapsulant disposed between the first redistribution structure and the second redistribution structure; second conductive vias extending through the encapsulant, the second conductive vias connected to the second metallization patterns of the second redistribution structure and the first metallization patterns of the first redistribution structure; a memory package; and second conductive connectors connecting the memory package to the second metallization patterns of the second redistribution structure.

14. The device of claim 10, wherein the active devices of the processor die are complementary metal-oxide-semiconductor devices, and the power semiconductor devices of the power gating die are insulated-gate bipolar transistor devices or double diffused metal-oxide-semiconductor devices.

15. The device of claim 10, wherein the processor die and the power gating die are directly bonded in a face-to-face manner.

16. The device of claim 10, wherein the processor die and the power gating die are directly bonded in a face-to-back manner.

17. A device comprising: a processor die comprising circuit blocks, the circuit blocks comprising active devices, the active devices being complementary metal-oxide-semiconductor devices; a power gating die connected to a first subset of the circuit blocks, the power gating die comprising power semiconductor devices, the power semiconductor devices being insulated-gate bipolar transistor devices or double diffused metal-oxide-semiconductor devices, wherein the power gating die is configured to receive a control signal from the processor die, and to turn one of the first subset of the circuit blocks of the processor die on or off responsive to the control signal; a dielectric layer around the power gating die; conductive vias extending through the dielectric layer, the conductive vias connected to a second subset of the circuit blocks; and a redistribution structure on the dielectric layer and the power gating die, the redistribution structure comprising metallization patterns, the metallization patterns connected to the conductive vias and to the power gating die.

18. The device of claim 17, wherein the redistribution structure, the dielectric layer, and the processor die are laterally coterminous.

19. The device of claim 17 further comprising: an encapsulant around the processor die and the dielectric layer.

20. The device of claim 17, wherein the active devices of the processor die and the power semiconductor devices of the power gating die have different pitches.
